Practical 5b: Product Table with EF Core (Insert Operation)

AIM: To create a Product table in SQL Server and perform Insert Operation using Entity Framework Core in an ASP.NET Core application.

Steps

1. Install NuGet packages: Microsoft.EntityFrameworkCore, Microsoft.EntityFrameworkCore.SqlServer, Microsoft.EntityFrameworkCore.Tools
2. Create the Product model and DbContext
3. Configure connection string in appsettings.json
4. Run migrations: Add-Migration Init → Update-Database

Product.cs

```
using System.ComponentModel.DataAnnotations;
namespace WebApplication5.Models
{
   public class Product
   {
      [Key]
      public int Id { get; set; }

      [Required]
      public string Name { get; set; } = string.Empty;

      public decimal Price { get; set; }

      public int Quantity { get; set; }
   }
}
```

AppDbContext.cs

```
using Microsoft.EntityFrameworkCore;
using WebApplication5.Models;
namespace WebApplication5.Models.Data
{
   public class AppDbContext : DbContext
   {
      public AppDbContext(DbContextOptions<AppDbContext> options)
         : base(options)
      {
      }
```

```
        public DbSet<Product> Products { get; set; }
    }
}
```

ProductController.cs

```
using Microsoft.AspNetCore.Mvc;
using WebApplication5.Models;
using WebApplication5.Models.Data;

namespace WebApplication5.Controllers
{
    public class ProductController : Controller
    {
        private readonly AppDbContext _context;

        public ProductController(AppDbContext context)
        {
            _context = context;
        }

        // Display Form
        public IActionResult Index()
        {
            return View();
        }

        // Insert Product
        [HttpPost]
        public IActionResult Index(Product product)
        {
            if (ModelState.IsValid)
            {
                _context.Products.Add(product);
                _context.SaveChanges();

                ViewBag.Message = "Product Inserted Successfully";
                ModelState.Clear();
            }

            return View();
        }
    }
}
```

Appsetting Json file

```
"ConnectionStrings": {
  "DefaultConnection":
"Server=(localdb)\\MSSQLLocalDB;Database=ProductDB;Trusted_Connection=True;Multiple
ActiveResultSets=true;TrustServerCertificate=True"
```

Program.cs

```
using Microsoft.EntityFrameworkCore;
using WebApplication5.Models.Data;

var builder = WebApplication.CreateBuilder(args);

builder.Services.AddControllersWithViews();

builder.Services.AddDbContext<AppDbContext>(options =>
    options.UseSqlServer(
        builder.Configuration.GetConnectionString("DefaultConnection")));

var app = builder.Build();

app.UseHttpsRedirection();
app.UseStaticFiles();

app.UseRouting();

app.UseAuthorization();

app.MapControllerRoute(
    name: "default",
    pattern: "{controller=Home}/{action=Index}/{id?}");

app.Run();
```

Index.cshtml

```
@model WebApplication5.Models.Product
@{
    ViewData["Title"] = "Product";
}
<h2>Insert Product</h2>

<form asp-action="Index" method="post">
```

```
    <div>
        <label asp-for="Name"></label>
        <input asp-for="Name" class="form-control" />
    </div>

    <br />

    <div>
        <label asp-for="Price"></label>
        <input asp-for="Price" class="form-control" />
    </div>

    <br />

    <div>
        <label asp-for="Quantity"></label>
        <input asp-for="Quantity" class="form-control" />
    </div>

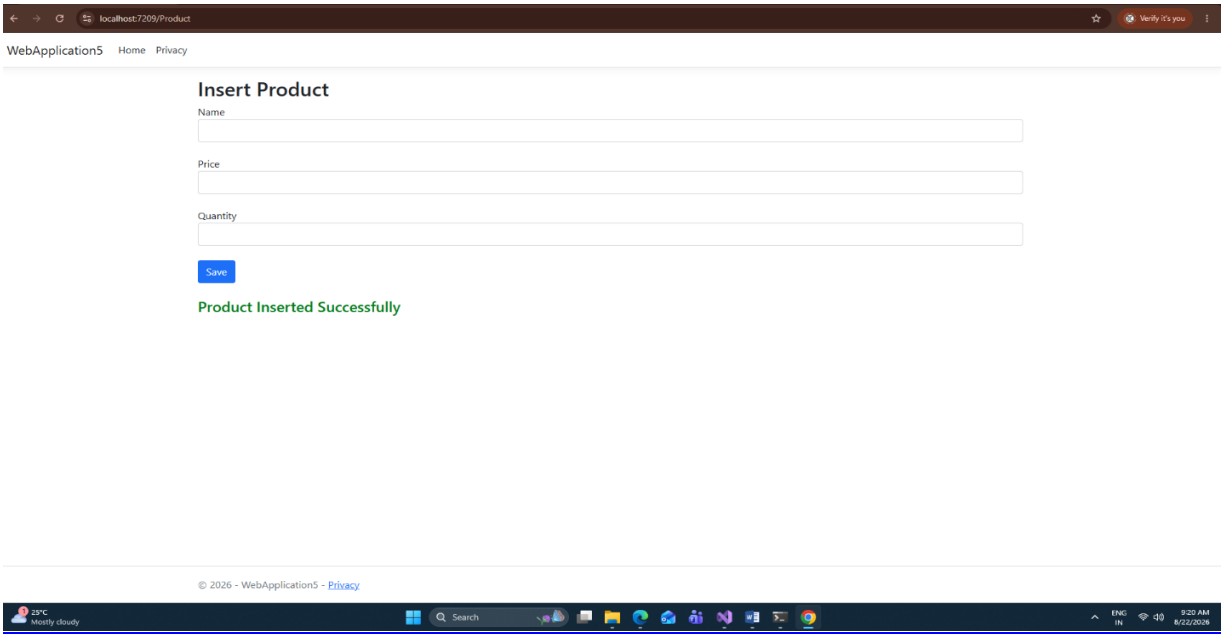
    <br />

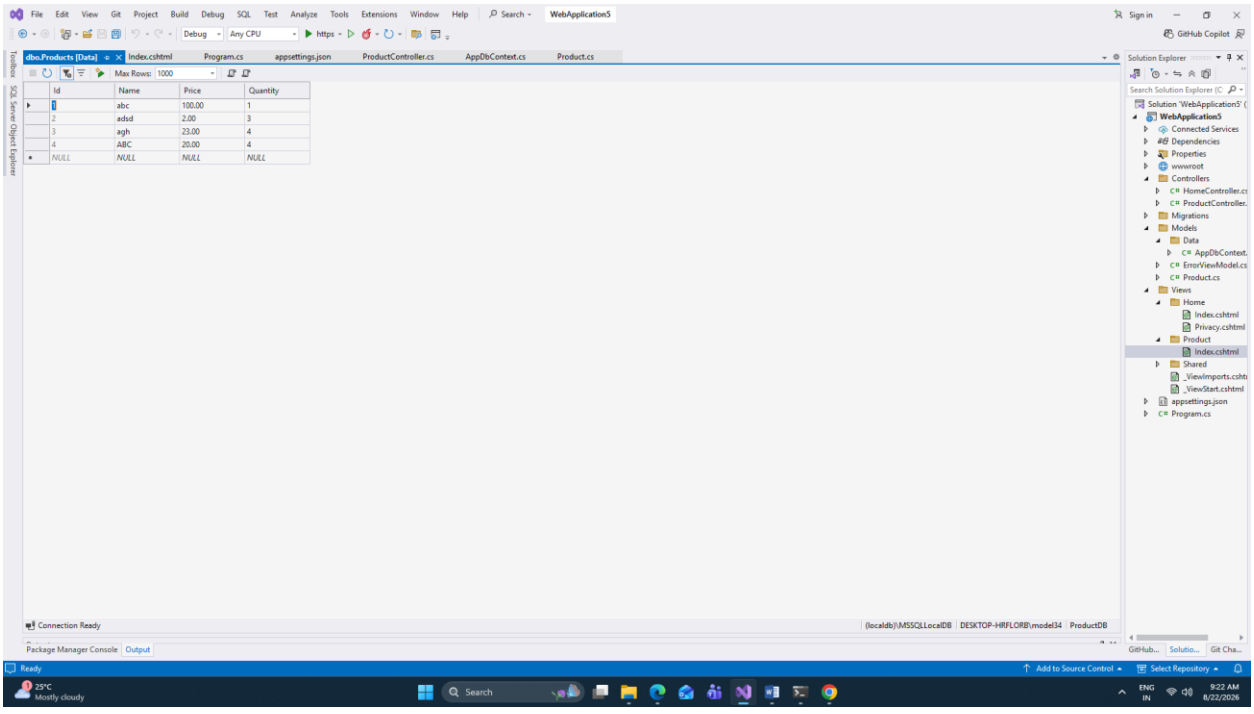
    <button type="submit" class="btn btn-primary">
        Save
    </button>

</form>

@if (ViewBag.Message != null)
{
    <br />
    <h4 style="color:green">@ViewBag.Message</h4>
}
```

Output


localhost:7209/Product
WebApplication5
Home
Privacy
Insert Product
Name
Price
Quantity
Save
Product Inserted Successfully
© 2026 - WebApplication5 - Privacy


WebApplication5
dbo.Products [Data]
Index.cshtml
Program.cs
appsettings.json
ProductController.cs
AppDbContext.cs
Product.cs
Id
Name
Price
Quantity
abc
adsd
agh
ABC
Connection Ready
Package Manager Console
Output
Ready